

Cute TEMPERATURE SENSOR

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Modern electronics have evolved to a great extent today, encompassing computer science on one hand and information technology on the other. There is hardly any electronic project nowadays that does not involve the use of a microcontroller unit (MCU) or computer programming languages.

However, there are still projects that can achieve the same goals as those requiring MCUs and software, but with simpler and more efficient designs.

In our daily lives, temperature controllers find numerous applications, such as in rice cookers, fridges, water geysers, microwave ovens, and air-conditioners. One device that stands out for its ability

to function as a splendid temperature controller, when paired with an MCU and suitable programming, is the TMP36. By precisely measuring temperatures and providing input to the MCU, which then converts the signal into control commands for associated devices like relays, the TMP36 simplifies temperature control. In this project, the TMP36 controls a relay directly, without the need for an MCU or software. The circuit, composed of a few discrete components, is both precise and straightforward. Fig. 1 shows the prototype tested in EFY lab.

Circuit and working

The circuit diagram of the cute temperature sensor is shown in

Fig. 2. The sensor is built around 5V voltage regulator LM7805 (IC1), the TMP36 sensor (IC2), a 5V one-changeover relay (RL1), npn transistor BC547 (T1), pnp transistor SK100 (T2), diode 1N60 (D1), and a few other components.

The core of the circuit is the TMP36 temperature sensor. It possesses the following characteristics: low voltage operation (2.7V to 5.5V), an operating temperature range of -40°C to $+150^{\circ}\text{C}$, linear operation ($y = mx + c$), temperature measurement directly in millivolts, uniform linearity of $20\text{mV}/^{\circ}\text{C}$, and a quiescent current of less than $50\mu\text{A}$. It also features a ramp up of $3^{\circ}\text{C}/\text{second}$ and a ramp down of $6^{\circ}\text{C}/\text{second}$. Pin configurations of TMP36 are shown in Fig. 3.

The project's working principle is simple. The TMP36 sensor detects the temperature and converts it into millivolts, which are then measured. When the TMP36 senses heat, the base voltage of transistor T1 exceeds the bias limit. As a result, both transistors T1 and T2 conduct, energising the 5V relay, which enables the load

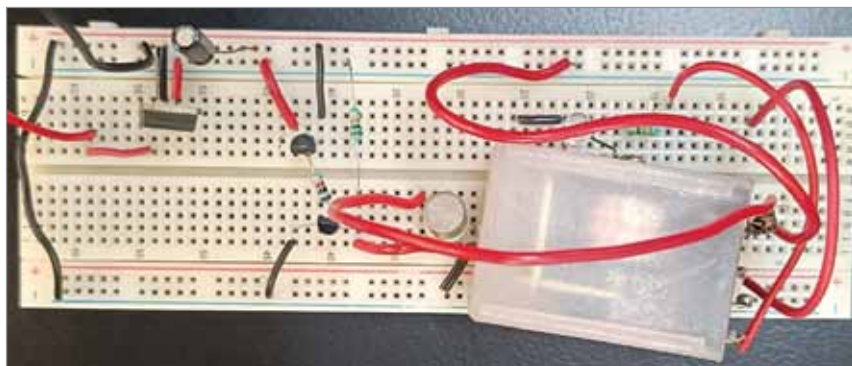


Fig. 1: Prototype tested in EFY lab

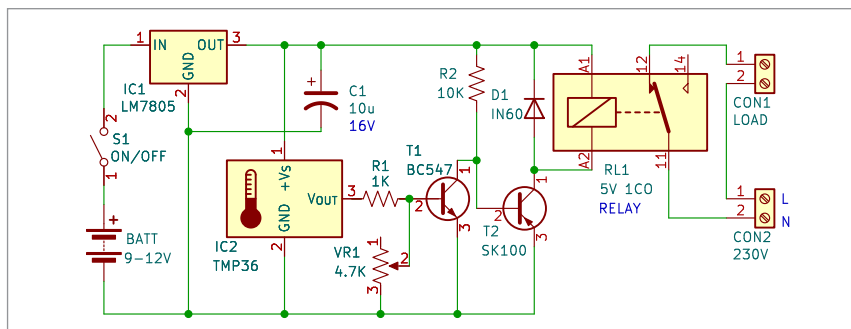


Fig 2: Circuit diagram

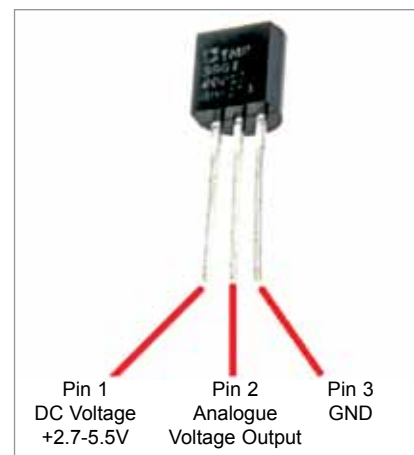


Fig. 3: TMP36 pins configuration